

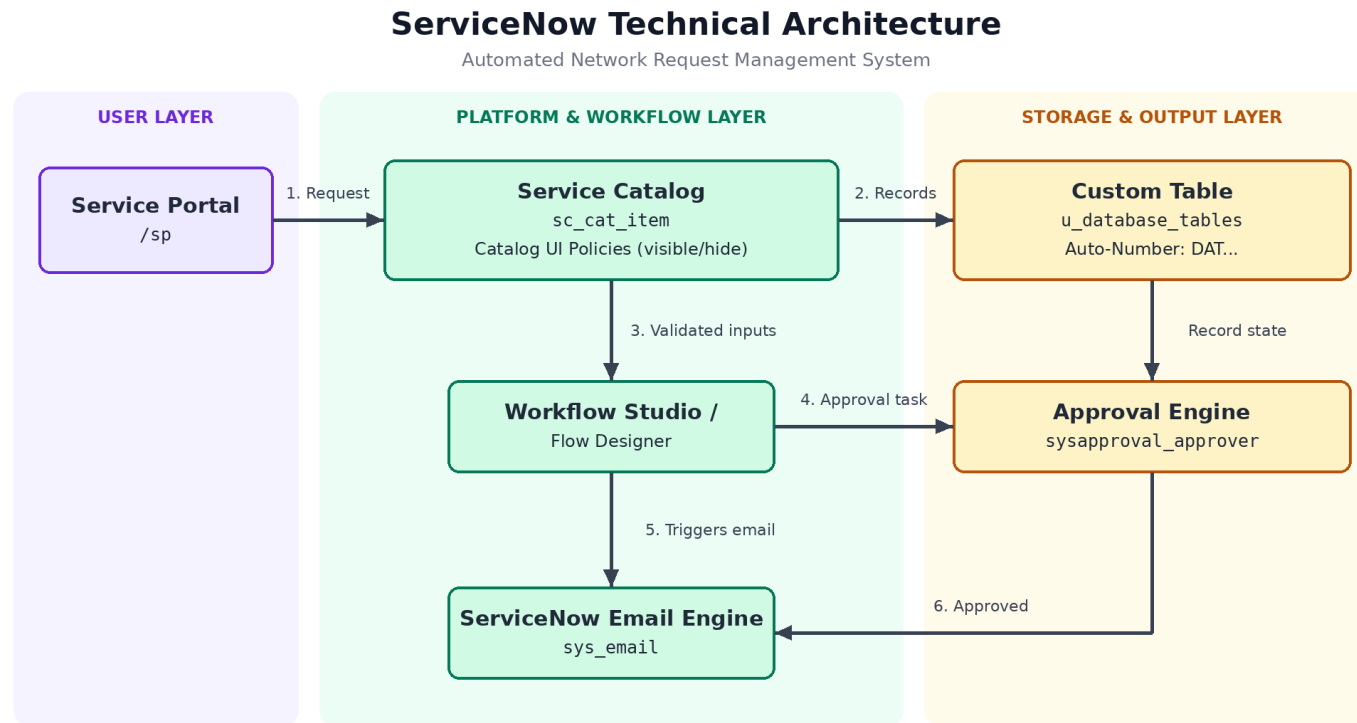
**Project Design Phase-II
Technology Stack (Architecture & Stack)**

Date	October 3, 2026
Team ID	
Project Name	Automated Network Request Management System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Automated Network Request Management System: ServiceNow Technical Architecture



Architecture Flow Steps:

User Interaction: End users access the Network Request catalog item via Service Portal.

Client Validation: Catalog UI Policies (visible/hide of ID field) dynamically manage mandatory fields and visibility prior to submission.

Flow Execution: Workflow Studio processes variables, inserts mapped records into custom table u_database_tables (DAT... prefix), and generates approval tasks.

Approval & Notification: Upon approval, Flow Designer updates the record state and triggers email output via ServiceNow's Email Engine (sys_email).

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Service Catalog portal for users to select options and submit network requests.	Service Portal, Service Catalog (sc_cat_item), HTML5, CSS3, Angular/Bootstrap Framework
2.	Application Logic-1	Client-side validation and dynamic field visibility rules.	ServiceNow Catalog UI Policies (visible/hide of ID field), Client Scripts
3.	Application Logic-2	Automated process orchestration, variable extraction, and state transitions.	ServiceNow Workflow Studio / Flow Designer
4.	Application Logic-3	Approval engine routing requests to designated managers/approvers.	ServiceNow Approval Engine (sysapproval_approver)
5.	Database	Platform database hosting custom tables and system records.	ServiceNow Relational Database Engine (u_database_tables)
6.	Cloud Database	Cloud-managed multi-instance platform architecture.	ServiceNow Cloud Infrastructure (PDI / Enterprise Cloud)
7.	File Storage	System logs, email notifications, and attachment handling.	ServiceNow System Logs, sys_email Table Engine
8.	External API-1	System notification engine dispatching dynamic email updates.	ServiceNow Email Engine / SMTP Service (sys_email)
9.	External API-2	Platform REST/SOAP APIs for external integrations and data exchange.	ServiceNow Table API / REST API
10.	Machine Learning Model	Predictive intelligence for auto-routing requests (Optional Platform Module).	ServiceNow Predictive Intelligence / AI Search
11.	Infrastructure (Server/Cloud)	Enterprise SaaS Cloud Instance deployment.	ServiceNow Cloud Infrastructure / Instance Platform (Utah / Washington DC)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Web components and rendering libraries supporting the portal interface.	AngularJS, Bootstrap, HTML5, JavaScript Client APIs
2.	Security Implementations	Role-based access control (RBAC), ACLs on custom table u_database_tables, HTTPS encryption.	Role-Based Access Control (RBAC), ACLs, HTTPS/TLS 1.3, Instance Security Center
3.	Scalable Architecture	Enterprise SaaS multi-tier cloud platform with separation of presentation, workflow, and data layers.	ServiceNow Advanced High Availability (AHA) Cloud Architecture
4.	Availability	Multi-instance cloud deployment with automated failover and zero single points of failure.	ServiceNow Cloud Cluster Load Balancers & Replication Services
5.	Performance	Asynchronous workflow execution via Flow Designer with optimized database indexing on auto-numbered records (DAT...).	ServiceNow In-Memory Caching, Flow Execution Engine

References:

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